GRACE VISION

Explainable OpenCV photo triage for animal adoption

A bounded workflow that measures cat-photo quality with OpenCV 5, proposes only conservative corrections, and requires a person to approve any selected image.

Executive summary

Animal shelters depend on trustworthy photographs, yet uploaded images may be dark, blurred, poorly framed, or missing reliable visual information. Generic generative editing is unsuitable where coat markings, body shape, injuries, and other identifying characteristics must remain authentic.

Grace Vision preserves each original, calculates visible evidence with OpenCV 5.0.0, chooses a bounded next action, optionally creates a separate deterministic derivative, measures it again, and returns the result to a human reviewer. It has no autonomous publishing path.

Core vision	Cloud execution	Human control
OpenCV 5.0.0 measurements and OpenCV DNN cat-framing evidence	AWS Lambda performs the material analysis; private S3 stores originals and derivatives	Explicit visibility review before the demo's selected image can change

Scope

- Competition copy only; GraceWays Animals production records and publishing remain disconnected.
- One to five rights-cleared JPEG/PNG images, up to 2 MiB and 1600 x 1600 pixels each.
- No inpainting, generative reconstruction, coat alteration, feature synthesis, or body reshaping.
- Evaluation evidence is deliberately labelled as synthetic control or rights-cleared real-photo evidence.

Technical report - 26 October 2026

1. User workflow and safety contract

Step	System behavior	Trust property
1	Reviewer selects 1-5 rights-cleared images.	No background ingestion.
2	Original bytes and SHA-256 are preserved.	Original is never overwritten.
3	OpenCV measures technical quality and cat framing.	Visible numeric evidence.
4	Policy selects one bounded action.	No open-ended autonomous editing.
5	A safe derivative is created only when warranted.	Deterministic, limited correction.
6	Derivative is measured with the same metrics.	Proposal is verified, not assumed better.
7	Reviewer sees comparison, reasons and trace.	Decision is explainable.
8	Reviewer approves or rejects explicitly.	No automatic publication.

Four allowed system actions

Action	Meaning
ACCEPT_ORIGINAL	Technical quality and framing pass; present the unchanged original for review.
ENHANCE	A limited correction is safe; create a separate derivative and remeasure it.
TRY_NEXT_IMAGE	Another candidate in the submitted set should be evaluated.
REQUEST_NEW_PHOTO	Reliable visual information is missing or the photo is unusable; do not reconstruct it.

Safety contract

- The original remains available and hash-identifiable throughout the workflow.
- Any derivative is stored separately and is never represented as new visual evidence.
- Missing detail triggers a new-photo request rather than synthesis.
- The reviewer sees the measurements and reasons before deciding.
- Approval changes only the demo selection; it does not publish or update a production animal record.

Human review is part of the safety architecture, not a cosmetic confirmation.

2. OpenCV 5 analysis and conservative processing

OpenCV supplies both the evidence and the post-processing verification. The recommendation is not a language-model assertion.

Resolution	Sharpness	Exposure
Pixel dimensions and minimum usable size	Laplacian variance	Mean brightness plus dark/clipped shares
Contrast	**Cat framing**	**Remeasurement**
Luminance spread	Confidence, box area, center and edge contact	The same metrics after any derivative

Cat-framing evidence

A pinned OpenCV Zoo YOLOX-S model runs through OpenCV DNN on CPU. It provides confidence and a bounding box. Grace Vision derives relative box area, center offset, and edge contact from that box. These signals guide framing policy, but they do not prove identity, anatomy, injury state, occlusion, or complete visibility.

Conservative corrections

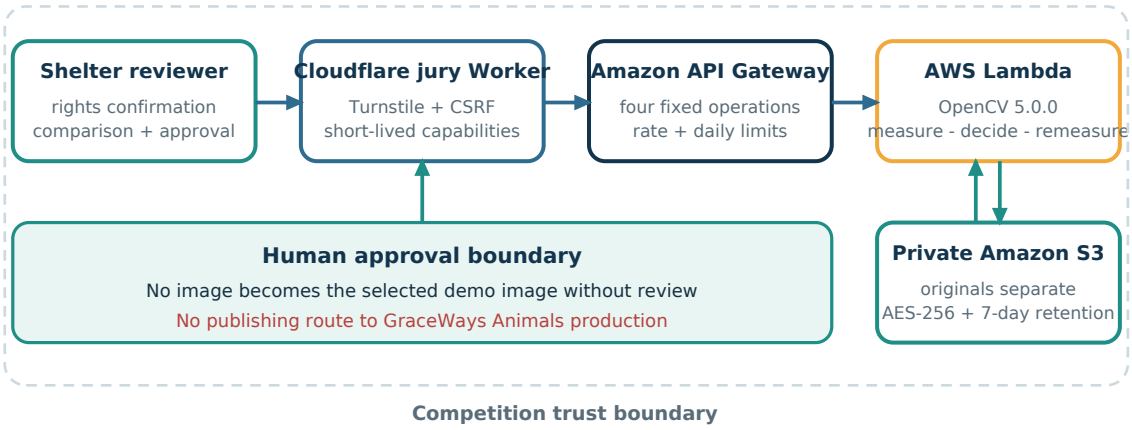
Operation	Boundary
Brightness	Gamma-based adjustment; no synthesis.
Contrast	Limited tonal adjustment.
Denoising	Conservative noise reduction.
Sharpening	Limited enhancement only; severe motion blur is rejected.
Cropping	Only when geometry is considered safe; otherwise preserve framing or request another photo.

Verification after processing

The derivative is decoded and measured again using the same quality functions. Before-and-after measurements and separate hashes are returned to the reviewer. This makes a claimed improvement inspectable and keeps the original as the authoritative source.

No hallucinated detail: if the input lacks reliable information, Grace Vision stops and asks for a new photo.

3. Architecture and material AWS contribution



Component	Responsibility
Browser	Rights confirmation, upload selection, evidence display, comparison and explicit decision.
Cloudflare jury Worker	Public security boundary; Turnstile, CSRF, short-lived session/job capabilities; keeps AWS credentials and storage details server-side.
Amazon API Gateway	Exposes only create, analyze, result and decide; API key, narrow Lambda permission, rate 2/burst 2 and 40 calls/day.
AWS Lambda container	Material OpenCV 5.0.0 decoding, measurement, DNN inference, policy, correction and remeasurement; reserved concurrency 2.
Private Amazon S3	Separate originals, derivatives, hashes and job state; public access blocked, AES-256, prefix-based seven-day retention.

The AWS component is substantive: the deployed Lambda container executes the OpenCV workload used by the jury interface. A complete reviewed job follows create - analyze - result - decide. The USD 5 monthly budget provides warnings only; it is not described as a hard cost cap.

4. Agentic Vision with bounded human control



Every branch records evidence, reasons and the next bounded action

Grace Vision qualifies as a bounded perception-decision-action workflow because measured visual state changes the subsequent action. The trace records the measurements, reasons, selected branch, optional transformation, remeasurement, and human decision. A chatbot that merely explains a fixed result would not provide this behavior.

Observed evidence	Selected next action	Human control
Quality and framing pass	ACCEPT_ORIGINAL	Review unchanged original.
Correctable technical issue	ENHANCE and remeasure	Compare original and proposal.
A better candidate remains	TRY_NEXT_IMAGE	Review the eventual best candidate.
Missing or unusable information	REQUEST_NEW_PHOTO	Provide a new source image.

Failure handling and observability

- No silent fallback from failed AWS processing to a mock success result.
- Unknown statuses receive a neutral message rather than a fabricated interpretation.
- One-time decisions and short-lived capabilities constrain repeated actions.
- A decision trace makes every branch visible to the reviewer and demonstrable to judges.
- The workflow can stop safely without changing or publishing the original.

Autonomy ends at a proposal. Authority remains with the reviewer.

5. Evaluation and demonstrated results

Evaluation evidence is separated by evidence type. The small real-photo sample is not presented as an accuracy benchmark.

Evidence	Observed result	Interpretation
Synthetic suitable control on AWS	ACCEPT_ORIGINAL; no derivative	Pass path and original preservation.
Synthetic correctable control on AWS	ENHANCE; derivative measured again	Correction path and verification.
Synthetic unusable control on AWS	REQUEST_NEW_PHOTO; no reconstruction	Safe rejection path.
Real-cat Molly boundary case	REQUEST_NEW_PHOTO despite technically correctable darkness/blur	Known false rejection caused by strict edge-contact policy.

Recorded correctable control

Metric	Original	Derivative	Observation
Resolution	800 x 640	800 x 640	Unchanged
Brightness	82.132	94.306	Increased
Sharpness	7610.638	8299.301	Increased
Contrast	43.468	45.392	Increased

Real-cat boundary case

Molly was detected with 86.8% confidence and a bounding box covering 77.9% of the frame. The image had technically correctable darkness and slight blur, but a strict edge-contact rule selected REQUEST_NEW_PHOTO. No derivative was approved and nothing was published. This is reported as a false rejection and a calibration target, not hidden as a success.

Engineering verification

- Phase 4H: 358 tests and 437 browser assertions passed.
- Later local session-renewal change: 107 targeted adapter, Worker and security checks passed; not claimed as deployed.
- Original and derivative SHA-256 values were confirmed through the demonstrated AWS workflow.

6. Security, reproducibility and responsible operation

Control	Verified design intent
Data separation	Originals and derivatives use separate objects and hashes; the original is never overwritten.
Storage	Private S3, public access blocked, AES-256, prefix-based seven-day lifecycle.
Browser boundary	AWS API key and signed storage details are not exposed to the browser.
Request integrity	Turnstile, CSRF, secure session handling and short-lived job capabilities.
Blast-radius limits	Lambda concurrency 2; gateway rate 2/burst 2; 40 operations/day.
Production isolation	No route to GraceWays Animals records or publishing.

Reproducibility

- OpenCV 5.0.0 is fixed in the container build.
- The YOLOX-S model is pinned by revision and SHA-256 in the build artifacts.
- The deployed container is referenced by an immutable image digest.
- Infrastructure is described as code and accompanied by build, deployment and test instructions.
- Synthetic controls provide deterministic pass, enhance and reject traces.

Container scan disclosure

The release scan reported 0 Critical and 7 High findings. The High findings were covered by a documented, time-limited manufacturer 'Pending Fix' risk acceptance. Grace Vision is not described as vulnerability-free.

Responsible-use commitments

- Use only images for which the uploader has appropriate rights.
- Treat cat detection as evidence, not identity or medical assessment.
- Never infer that a bounding box proves full anatomy or unchanged coat markings.
- Never auto-publish; keep the final decision with a responsible person.
- Minimize retention and disclose limits, failure cases and security assumptions.

7. Limitations, lessons and next steps

Known limitation	Consequence
Small rights-cleared real-photo evaluation	No population-level accuracy claim.
Bounding-box detector, not segmentation or identity verification	Cannot certify anatomy, occlusion, injuries or full visibility.
Strict edge-contact policy	May reject otherwise useful photos, as seen with Molly.
Input limits	JPEG/PNG, 2 MiB, 1600 x 1600 pixels; intentional demo boundary.
Short-lived public session	Fresh reload is required for a reliable jury run until the locally tested renewal change is deployed.
Competition isolation	No persistence into productive GraceWays records and no external publication.

Lessons

- Measurements, hashes and traces make assistance inspectable rather than magical.
- Conservative failure avoids fabricated detail but still requires calibration against real workflows.
- Human approval belongs in the technical architecture, not only in interface copy.

Next steps

- Expand a varied, rights-cleared real-cat evaluation set.
- Refine edge-contact handling so unsafe cropping can be blocked without blocking safe global correction.
- Validate thresholds with animal-welfare users.
- Deploy and verify the locally tested session-renewal update.
- Add optional authenticated GraceWays Animals integration only after a separate production security review.

Evidence status

Deployed and verified on AWS: OpenCV 5 execution, accept/enhance/reject control paths, private storage behavior, hashes and the human decision workflow. Verified locally: the later session-renewal change. Known limitation: Molly false rejection. Planned: broader real-photo evaluation and productive integration.

References

- OpenCV AI Competition 2026 overview and final submission requirements: https://opencv26.devpost.com/
- OpenCV AI Competition 2026 official rules and judging rubric: https://opencv26.devpost.com/rules
- Public jury demo: https://gracevision-opencv26-jury-demo.black-term-967a.workers.dev/

Grace Vision gives shelters measurable evidence, a conservative proposal, and the final say while preserving the original and avoiding generative reconstruction.